Metaplectic Jacquet–Langlands Transfer, Companion Paper (Rank Two, Wild Place)
Revision Log: Draft 5 to Draft 6

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Scope

Draft 6 responds to ChatGPT's referee report on Draft 5 (five recommendations). The main effect is that Theorem `thm:leveltwo` no longer depends on any new computation: both of its computational steps now have hand proofs. Section 12.5 is rewritten. Nothing outside Section 12, the abstract, Appendix B and one line of Section 1 changed.

Changes

1. **Step 2 by hand** (referee's §3.1, adopted verbatim in substance). $H = Q_3/Q_7$ has $H' = Q_6/Q_7$ of order 3 and center Q_4/Q_7 of index 9, so it has 243 linear and 54 degree-three characters, 27 above each nontrivial ω of H', and the isotypic decomposition of the regular character gives $\sum_{\chi'|_{H'}=3\omega} \chi' = 81\,\omega\,\mathbf{1}_{H'}$. Since χ_3 vanishes off the center of the Heisenberg group, $\chi_3 \circ r$ vanishes on the depth-three good classes, and on H' equals $3\,\omega_3 \circ r$ with $D = 27$; hence $(\chi_3 \circ r)D = \sum_{\chi'|_{H'}=3\omega_3\circ r} \chi'$, and the two halves add up to the sum of all 54. Script 14 is now corroboration.

2. **Step 3 by hand** (new Proposition `prop:free`). $\mathrm{Ad}(Q_1)$ acts freely on *all* genuine irreducible characters of $\hat{H} = \tilde{U}_3/\ell(K^7)\ker\tilde{\theta}_1$. Proof: the preimage $\hat{A}$ of $A = Q_4/Q_7$ is central in $\hat{H}$ (by $\omega = 0$, Lemma `lem:H1consequences`(c)), so a genuine irreducible μ has a central character $\hat{\nu} = \lambda_0\alpha$ on $\hat{A}$, and $\hat{\nu} \circ \mathrm{Ad}(t)/\hat{\nu} = c_t \cdot \omega_\mu([t,\cdot])$ with $c_t = \lambda_0^t/\lambda_0 = \zeta^{h_t}$, $h_t \in \Lambda$ nonzero for $t \notin Q_3$. For $t \in Q_2$, $[t, A] \subset K^7$ and the ratio is $c_t|_A$, nontrivial because no nonzero element of Λ vanishes on the Q_4/Q_6-coordinates (table in Appendix A). For t of depth one with leading coefficient a, on $y \in Q_5$ with leading coefficient b the ratio is $\zeta^{2\beta\mathrm{Re}(a\bar{b})+2\gamma\mathrm{Im}(a\bar{b})}$ ($\beta \neq 0$ from Lemma `lem:H1consequences`(b); the commutator class $2i\,\mathrm{Im}(a\bar{b})$ was also checked numerically), a nonzero functional of $a\bar{b}$. Hence no genuine μ is fixed by any $t \notin Q_3$. Script 15 is now corroboration, and writes the certificate the referee asked for (`level2_orbit_certificate.txt`: full action tables of the 27 coset representatives on the 54 degree-three and 243 linear genuine characters, explicit stabilizer test with zero fixed pairs, orbit lists).

3. **Normalization** (referee's §3.3). Section 12 now fixes $\theta_1 := \mathbf{1}$ throughout (allowed by Remark `rem:theta-independence`; it is the scripts' normalization), so $\theta_0 = \mathbf{1}$ or sgn, and λ_0, $\hat{\Gamma}$, $\hat{H}$ are all defined in the same normalization; the parenthetical in Step 1 says how $\ker\tilde{\theta}_1$ and ε_ρ are used when $\theta_\rho = \mathrm{sgn}$.

4. **Section 12.5 rewritten** (referee's §4). (a) The cube root is a surjection from the good classes of H_m onto Q_1/Q_{m+1}, many-to-one ($219 \to 27$ for $m = 2$), not a bijection. (b) $\hat{H}_m$ is defined with $K^{\max(3m+1,6)}$; for $m = 1$ it is $\tilde{V}$, whose 243 genuine characters form nine free orbits. (c) The constituent-wise "cube-root correspondence" is withdrawn: the referee's example ($\chi(a + bi) = \zeta^a$, $w = 1 + i$, $w^5 = -w$) shows $\chi \circ r \neq \chi' \circ \pi_3$ for every χ', and $\Phi_1(\chi \circ r)$ is not a virtual character for a single additive $\chi \neq 1$. The correct statements are for D_8-invariant characters (restrictions of characters of $D^\times$), which is all Target B needs; at level two the constituent-wise identity holds for χ_3 because χ_3 vanishes off the center of the Heisenberg group. The section now states the two ingredients (freeness; $(\Theta_\rho \circ r)D$ a character of H_m) that gave the level theorems, notes that both are open for $m \geq 3$, and sketches the inductive structure of a possible proof of freeness ($\hat{A}_m$ central in $\hat{H}_m$ for $m \geq 3$ without hypotheses, since ℓ is a homomorphism on U_6). The term "cube-root correspondence" is removed from the abstract and table.

5. **Provenance** (referee's §5). Appendix B now states that scripts 13–14 obtain the character table of Q_3/Q_7 numerically (Burnside) and verify it exactly, that orthogonality is a verification and not a certificate of irreducibility, and that since Draft 6 these scripts are corroboration only. The abstract now says "compatible with the cover (central character trivial on μ_3)". The introduction's draft range is updated.

Added after the referee round (same day)

At David Joyner's request Draft 6 also gains a new Section 2 ("Background"), with three subsections: division algebras over local and global fields (Hasse invariants, the local structure $D = \bigoplus E_0\Pi^k$, maximal subfields, the filtration K^m, the Albert–Brauer–Hasse–Noether theorem, Jacquet–Langlands); the cover, star map and good elements (a summary of what is imported from Part I); and Heisenberg groups, the Weil representation and the Howe correspondence (Stone–von Neumann, Schrödinger model, dual pairs, Howe duality, Waldspurger's realization of the Shimura correspondence through $(\widetilde{\mathrm{SL}}_2, O(V_3))$ with V_3 split or anisotropic, Howe's Kirillov theory), ending with the three points of contact with this paper. A new Remark `rem:howe-duality` in the interface section reformulates Target B as a Howe-duality statement: the kernel $K_\theta(x, g) = \sum_\rho L_D \Theta_\rho(x)\overline{\Theta_\rho(g)}$ equals $\Theta_{\tau_1}(x)\,\tilde{\theta}\tilde{\theta}_1^{-1}(\tilde{z})\,\Delta_\theta(t, g)$ on star lifts $x = \zeta\tilde{z}t^*$, and Target B for θ is equivalent to K_θ being the character of a representation $\Omega_\theta = \bigoplus_\rho \tau_\rho \otimes \rho^\vee$ of $\tilde{H} \times H$; this is labeled a reformulation, not a result. Thirteen references were added. Because of the new Section 2 all later section and statement numbers increase by one (e.g. the level-two theorem is now Theorem 13.7, freeness Proposition 13.8); the labels are unchanged and the scripts' README was updated accordingly.

Status

Theorem `thm:leveltwo`: [P^c], now depending on Target A only (through λ_0, Λ, e_*), with no new computational input. Proposition `prop:free`: [P^c] for the same reason. Section 12.5: [O]. Nothing else changed.